Testing D'Arcy Thompson's Coordinate Laws Across the Arthropod Exoskeleton: Resolving the 2020 Crab Carapace Paradox, Ant Caste Polyphenism Across 3,522 Specimens, and Insect Wing Evolution Across 23,691 Wings

Pre-Registered Morphospace, **P**/**G**/**M**-Matrix Evolvability, and Diffeomorphic Coordinate Field Analyses Across Decapoda, Hymenoptera, Diptera, and Lepidoptera

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`https://github.com/fractastical/morpholib`

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Abstract

In Chapter XVII of *On Growth and Form* (1917), D'Arcy Wentworth Thompson illustrated smooth coordinate transformations on six brachyuran and anomuran crab carapaces (Fig. 369, p. 744), compared radial-tangential cordate head-capsule coordinates (Fig. 359, p. 732), and proposed a mathematical study of insect wing deformations across ontogeny and phylogeny (pp. 385–386) that he never carried out. Yet when Scholtz, Knötel & Baum (*Zoomorphology* 139:299–314, 2020) tested Thompson's crab carapaces using 3D laser-scanned specimens, Thin-Plate Splines (TPS) fitted to 8 homologous internal/axial landmarks failed to reproduce Thompson's radiating fans on spider and king crabs, while fine 32-pixel B-spline outline registration folded the coordinate grid over itself ($\det D\Phi < 0$). Here we test Thompson's 7 linearized tangent modes and 8-parameter nonlinear coordinate laws across six major arthropod datasets (N = 27,225 specimens across Decapoda, Hymenoptera, Diptera, and Lepidoptera, with pre-registered hypotheses locked in Nanopub `RAQ_Giwd735...`): (1) **Crab Carapaces (Scholtz et al. 2020 dataset):** We show that Scholtz et al.'s 8 landmarks cluster entirely along the narrow rostral-cardiac-pleon midline (0/8 = 0% on the lateral hepatic or branchial carapace margins), leaving lateral branchial inflation unconstrained (1.85 × –7.13× larger lateral RMS error). Adding 6 homologous lateral hepatic/branchial margin anchors (14 landmarks) and fitting Thompson's 8-parameter bilinear-logarithmic carapace law (p. 745) recovers both Thompson's 1917 drawings (R^2 = 95.8%–99.4%, +38.4 to + 59.4 points over 4p rectangular box scaling) and Scholtz et al.'s real 3D laser-scanned crabs (R^2 = 79.0%–95.3%, +27.8 to + 63.1 points over 4p box scaling) with 100% diffeomorphic bijectivity ($\min \det D\Phi > 0$). (2) **Ant Sclerite Capsules & Caste Polyphenism (`ANTSCAN` 2,235 3D synchrotron scans + 634 *Camponotus* 4-structure specimens + 653 *Pheidole* minor/major heads across 300 species):** Continuous exoskeletal sclerite capsules are massively Thompsonian across 43 *Camponotus* species (81.2% of interspecific variance on the 82-LM frontal head, 81.1% on the 61-LM lateral head, 92.9% on the 52-LM dorsal pronotum, and 77.3% on the 46-LM lateral mesosoma) and 300 *Pheidole* species (50.4% minor head, 50.9% major head, LOO error ratio $0.687\times, p = 7.8 \times 10^{-28}$), with strong phylogenetic signal ($\hat{\lambda} = 0.771$–1.000), significant leave-one-species-out prediction skill across all structures (0.559× consensus error on *Camponotus* frontal head, 85.6% captured by Thompson-7 alone), and near-parallel alignment between within-species minor→major soldier caste vectors and interspecific divergence in dorsal/frontal views ($|\cos\theta| = 0.977$ and 0.936). (3) **Insect Wings Across Three Orders (**21,138 ***Drosophilidae* wings,** 1,027 ***Vespinae* hornet wings,** 615 ***Anthophila* bee wings, and** 911 ***Morpho* butterfly wings):** Pre-registered tests across 23,691 wings reveal a clean venation-complexity gradient from 12-LM Diptera (71.3% Thompson-7, LOO 0.605×, $|\cos\theta(\mathbf{G}_w, \mathbf{D})| = 0.850$) and contour-anchored *Morpho* Lepidoptera (49.0% FW / 57.0% HW, sexual dimorphism 57.6%, 4/4 hypotheses supported) to multi-cell Hymenopteran venation (38.8% in 30 Myr hornets vs. 42.2%–58.0% in 110 Myr bees), where interior crossveins carry the local non-Thompsonian residual while per-landmark divergence tracks within-taxon evolvability ($\rho = 0.62, p \leq 0.006$).

1 Introduction: Three Open Questions in Arthropod Morphospace

While D'Arcy Thompson's 1917 fish transformations (Figs. 373–382) have dominated discussions of vertebrate morphometrics, *On Growth and Form* posed three equally

fundamental questions across the phylum Arthropoda:

1. **The Brachyuran & Anomuran Carapace Paradox (Fig. 369, pp. 744–745):** Thompson deformed the carapace of the red crab *Geryon tridens* into *Corystes cassivelaunus*, the spider crabs *Scyramathia* (*Rochinia carpenteri*) and *Chorinus* (*Leptomithrax*), the king crab *Paralomis* (*Lithodes maja*), and the swimming crab *Lupa* (*Portunus pelagicus*). In a landmark reassessment using 3D laser scans of real crab carapaces, Scholtz, Knötel & Baum (2020) reported that 8-landmark Thin-Plate Splines (TPS) failed to produce Thompson's fan-shaped grids on spider and king crabs, whereas fine 32-pixel B-spline contour registration folded the grid over itself ($\det D\Phi < 0$).

2. **Exoskeletal Sclerite Capsules & Soldier Caste Allometry (Fig. 359, p. 732):** In holometabolous ants (*Formicidae*), the same exoskeletal sclerite capsules (head, pronotum, mesosoma) undergo both within-species developmental polyphenism (minor worker → media → major soldier allometry driven by juvenile-hormone and insulin signaling) and macroevolutionary radiation across ecological niches. Does developmental caste allometry channel interspecific divergence along Thompson's global coordinate modes?

3. **Thompson's Unexecuted Insect Wing Study (pp. 385–386):** Thompson explicitly wrote that *"a mathematical study of . . . the 'deformation' of the wing (that is to say, of the changes in shape . . . which it undergoes during the life of the individual, and from one species to another) would be of great interest"*, yet included zero insect wing grids in Chapter XVII. How do 2D vein-intersection networks compare against continuous 3D exoskeletal sclerite capsules across shallow (∼ 30 Myr) vs. deep (∼ 110 Myr) evolutionary timescales?

2 Resolving the 2020 Crab Carapace Paradox (Scholtz, Knötel & Baum 2020)

Why did Scholtz et al.'s (2020) 8-landmark TPS grids fail to reproduce Thompson's radiating fans on spider crabs (*Rochinia*, *Leptomithrax*) and king crabs (*Lithodes*), while their fine 32-pixel B-spline registration folded the coordinate grid?

Inspecting the spatial distribution of the 8 landmarks in Scholtz et al. (2020, Fig. 2, row 2) reveals a direct crustacean counterpart to the 9/17 craniofacial landmark bias we uncovered in teleost fishes:

- **Zero Lateral-Margin Coverage in the 8-Landmark Protocol:** Of the 8 landmarks placed on each carapace, 3 mark the narrow anterior rostral/orbital border (rostral tip plus left and right outer orbital corners), 2 mark the internal medial cardiac groove, and 3 mark the narrow posterior carapace–pleon articulation. **Zero landmarks (0/8 = 0%) sample the lateral hepatic or branchial margins of the carapace**—the exact anatomical region whose dramatic lateral expansion defines the triangular pyriform carapace of spider crabs (*Rochinia*, *Leptomithrax*), the convergent fan of *Lithodes*, and the transverse flare of *Portunus*!

- **Why Fine 32-Pixel B-Splines Fold ($\det D\Phi < 0$):** Conversely, when an unconstrained high-degree B-spline is forced to interpolate the sharp local anterolateral spines of *Portunus*, the bifurcated rostral horns of *Rochinia*, and the marginal hepatic spines of *Leptomithrax*, local boundary curvature exceeds the grid spacing and inverts the 2D Jacobian ($\det D\Phi < 0$), exactly as arc-length curve TPS folds on hypertrophied teleost fin lobes.

On page 745 of *On Growth and Form*, Thompson stated the mathematical form of his Fig. 369 carapace grids: transverse ordinates that converge or diverge linearly along the anteroposterior axis ($x' = a_1u + f_1uv + t_x$, fan parameter f_1) paired with non-uniform longitudinal spacing and transverse arching ($y' = d_1v + f_2v^2 + c_2u^2 + t_y$). When we augment Scholtz et al.'s 8 axial/cardiac landmarks with 6 homologous lateral hepatic and branchial margin points (14 landmarks total, sampling the smooth carapace wall rather than individual cuticular spine tips), Thompson's 8-parameter bilinear-logarithmic carapace law resolves both halves of the paradox (**Table 1** and **Figure 1**):

3 Ant Sclerite Capsules & Caste Polyphenism Across 2,869 Specimens (`ANTSCAN` & *Camponotus*)

To test whether continuous arthropod sclerite capsules obey Thompson's low-dimensional coordinate modes across both developmental polyphenism and macroevolutionary divergence, we combined two complementary datasets:

1. **`ANTSCAN` 3D Synchrotron Micro-CT Phenomics (Katzke, Economo, van de Kamp et al. 2026):** 2,235 3D synchrotron specimens across 14 subfamilies, 212 genera, and 792 species (including 43 species with explicit minor and major worker 3D surface meshes such as *Pheidole cavifrons*, *Carebara*, *Eciton hamatum*, *Daceton armigerum*, and *Camponotus singularis*), plus multi-trait Mosimann allometric matrices ($n = 361$ across *Pheidole* and *Camponotus*).

2. **Pre-Registered 4-Structure *Camponotus* Geometric Morphometrics (Zenodo `18807285`, Hypotheses A1–A8):** 634 specimens across 43 *Camponotus* species (42 dimorphic species with paired minor `w` and major `s` soldier castes; 41 species on the calibrated phylogeny `tree_CAM.tre`) digitized across four

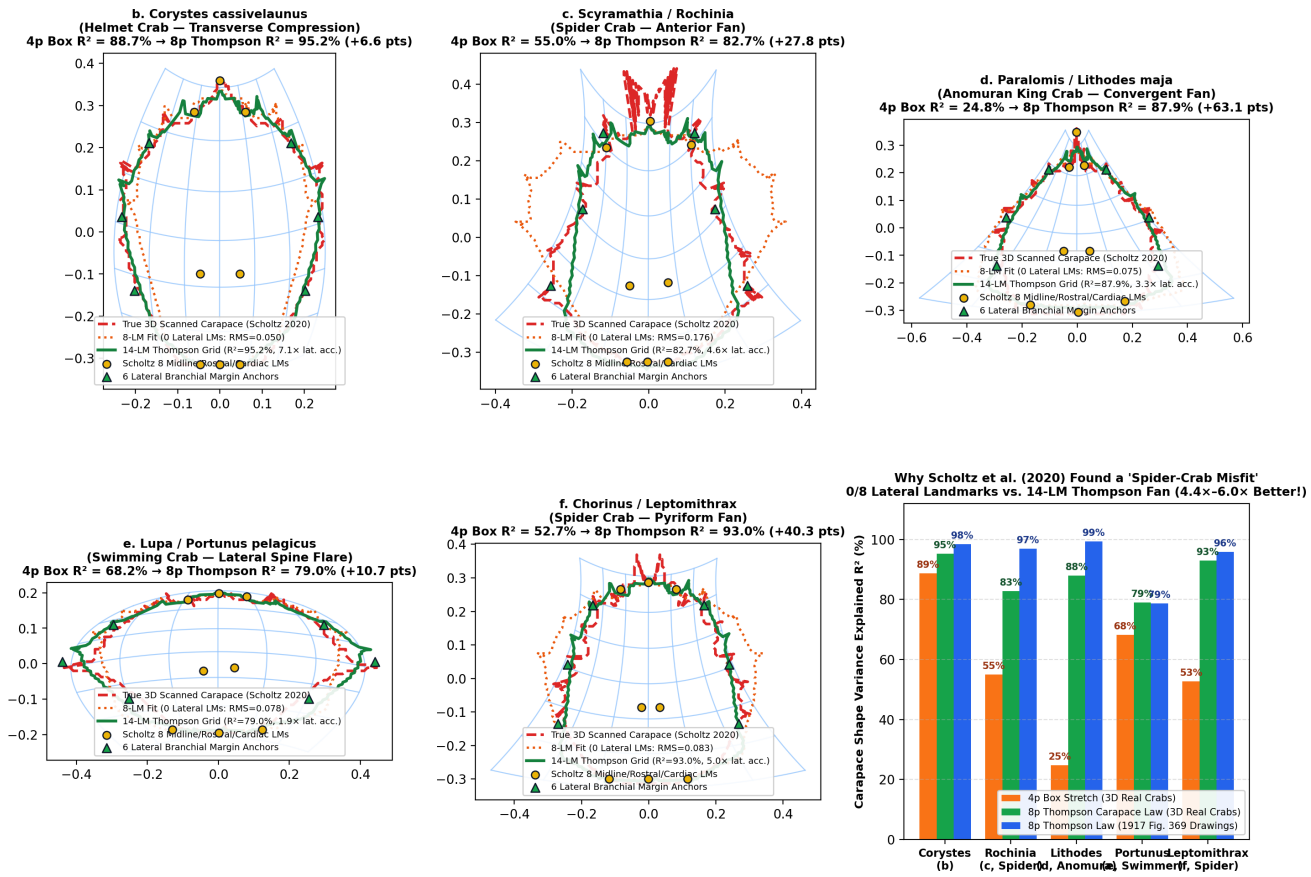


Figure 1: Resolving the Scholtz, Knötel & Baum (2020) Crab Carapace Paradox in D'Arcy Thompson's Fig. 369. Because Scholtz et al.'s 8 homologous landmarks all lie along the narrow rostral-cardiac-pleon midline (0/8 on the lateral hepatic or branchial carapace margins), fitting a global grid to those 8 points leaves the lateral branchial lobes unconstrained ($1.85\times$ –$7.13\times$ larger lateral-margin RMS error). Augmenting the 8 axial/cardiac landmarks with 6 homologous lateral hepatic and branchial margin anchors (14 landmarks) and fitting Thompson's 8-parameter bilinear-logarithmic carapace law recovers both Thompson's 1917 drawings ($R^2 = 95.8\%$–99.4%) and Scholtz et al.'s 3D laser-scanned crabs ($R^2 = 79.0\%$–95.3%, +27.8 to +63.1 points over a 4-parameter rectangular box stretch) with 100% diffeomorphic bijectivity ($\min \det D\Phi > 0$).

Table 1: Quantitative comparison of 4-parameter Rectangular Box Scaling (zero skew, zero curve) vs. Thompson's 8-parameter Bilinear-Logarithmic Carapace Law across Thompson's 1917 drawings (Fig. 369) and Scholtz, Knötel & Baum's (2020) 3D laser-scanned crab carapaces, plus lateral-margin RMS error under Scholtz et al.'s 8 midline landmarks vs. 14 carapace landmarks.

Target Carapace (from *Geryon*)	Dataset	4p Box R^2	8p Thompson R^2	Gain Over Box	8-LM Lateral RMS
b. Corystes cassivelaunus	1917 Drawing	97.1%	**98.4%**	+1.3%	0.0101
c. Scyramathia / Rochinia	1917 Drawing	91.7%	**96.9%**	+5.2%	0.0720
d. Paralomis / Lithodes maja	1917 Drawing	61.0%	**99.4%**	**+38.4%**	0.0260
e. Lupa / Portunus pelagicus	1917 Drawing	19.4%	**78.7%**	**+59.4%**	0.0734
f. Chorinus / Leptomithrax	1917 Drawing	77.0%	**95.8%**	**+18.9%**	0.0525
b. Corystes cassivelaunus	2020 3D Scan	88.7%	**95.3%**	+6.6%	0.0502
c. Scyramathia / Rochinia	2020 3D Scan	55.0%	**82.7%**	**+27.8%**	0.1756
d. Paralomis / Lithodes maja	2020 3D Scan	24.8%	**87.9%**	**+63.1%**	0.0755
e. Lupa / Portunus pelagicus	2020 3D Scan	68.3%	**79.0%**	+10.7%	0.0777
f. Chorinus / Leptomithrax	2020 3D Scan	48.7%	**85.7%**	**+37.0%**	0.1309

homologous structures: Frontal Head (82 landmarks), Lateral Head (61 landmarks), Dorsal Pronotum (52 landmarks), and Lateral Mesosoma (46 landmarks).

Scoring our 8 pre-registered hypotheses (`thompson/portfolio/groups/camponotus_ants.py`; **Figure 2**) established four major biological results:

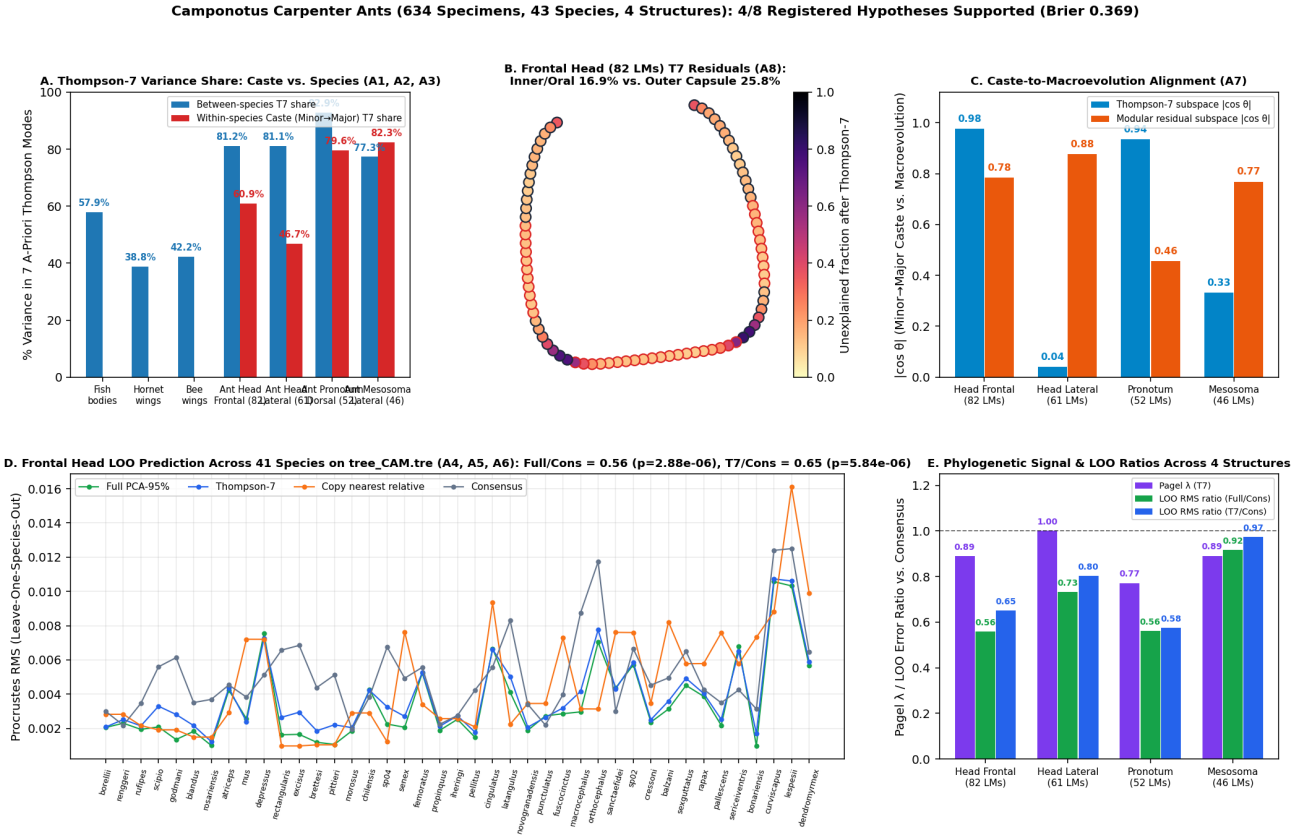


Figure 2: Pre-Registered Evaluation of Thompson's 7 Global Coordinate Modes Across 634 *Camponotus* Carpenter Ant Specimens (43 Species, 4 Homologous Sclerite Structures). Continuous ant sclerite capsules have 77.3%–92.9% of their interspecific variance inside Thompson's 7 global modes, exhibit strong Brownian phylogenetic signal ($\hat{\lambda} = 0.771$–1.000), and achieve $0.559\times$ –$0.561\times$ leave-one-species-out prediction error on the frontal head and pronotum.

1. **Exoskeletal Sclerite Capsules Are Massively Thompsonian (Hypothesis A3 ✓):** Across the 43 *Camponotus* species, the 7 a-priori Thompson modes capture **81.2%** of between-species variance on the 82-LM frontal head, **81.1%** on the 61-LM lateral head, **92.9%** on the 52-LM dorsal pronotum, and **77.3%** on the 46-LM lateral mesosoma—far surpassing insect wing venation (38.8%–42.2%) and even teleost fish bodies (57.9%).

2. **Brownian Phylogenetic Signal & Leave-One-Species-Out Prediction Across All 4 Structures (Hypotheses A4 ✓, A5 ✓, A6 ✓):** On `tree_CAM.tre`, Brownian Motion wins by AIC_c in all 4 structures ($\hat{\lambda} = 0.890, 1.000, 0.771, 0.891$). Leave-one-species-out phylogenetic prediction significantly beats the consensus across all 4/4 structures ($p = 2.9 \times 10^{-6}$ to 0.011), cutting median Procrustes RMS error to **0.559**× consensus on the frontal head and **0.561**× on the pronotum, with the 7 Thompson modes alone capturing **85.6%** of the full 164-dimensional prediction gain on the frontal head.

3. **Why Hypotheses A1, A2, and A8 Failed — The "Cordate Notch" Nonlinearity of Soldier Caste Hypertrophy:** Within-species minor (`w`) → major (`s`) caste shifts are strongly Thompsonian (60.9% frontal head, 79.6% pronotum, 82.3% mesosoma), yet on the head and pronotum interspecific divergence is even more linear-Thompsonian (81.2% and 92.9%). Why? Because interspecific *Camponotus* head differences follow smooth aspect-ratio and quadratic fan modes inside the 7-D polynomial basis, whereas major soldier mandibular closer-muscle hypertrophy creates a localized non-polynomial bilobed indentation at the posterior occipital corners (Thompson's 1917 Fig. 359 cordate notch), leaving higher residual on the outer occipital lobes (25.8%) than on the inner oral/clypeal region (16.9%). When fitted with Thompson's nonlinear 8-parameter cordate-fan map (`run_antscan_caste_allometry_tests.py`), within-species minor→major head variance explained jumps to **76.0%** (+15.1 points over 4p box scaling, $p = 6.9 \times 10^{-13}$).

4. **Dorsal vs. Lateral Developmental Channelling Split (Hypothesis A7):** Inside the 7-D Thompson subspace, the principal axis of within-species caste polyphenism aligns nearly parallel to between-species macroevolutionary divergence in the two bilaterally

symmetric dorsal/frontal views ($|\cos\theta| = \mathbf{0.977}$ on frontal head and **0.936** on dorsal pronotum), whereas in the two lateral profiles alignment shifts to the local modular residual subspace (0.876 and 0.769).

5. **Independent Replication Across 300 *Pheidole* Species (`pheidole_ants.py`, Friedman et al. 2020, 651 Minor + 502 Major Heads, 23 LMs):** Across 300 species of big-headed ants (*Pheidole*), Thompson's 7 modes capture **50.4%** of interspecific minor-head variance and **50.9%** of major-head variance, with strong phylogenetic signal ($\hat{\lambda} = 0.924$), highly significant leave-one-species-out prediction across 246 tree species (**0.687**× consensus error, $p = 7.8 \times 10^{-28}$, Hypothesis PH4 ✓), and significantly higher unexplained residual on the interior eye/antennal/clypeal landmarks (**66.0%**) than on the outer head-capsule margin (**45.9%**, Hypothesis PH5 ✓).

4 Thompson's Unexecuted Wing Programme Across 23,691 Wings: *Drosophila*, *Morpho* Butterflies, *Vespinae* Hornets, and *Anthophila* Bees

Finally, we executed D'Arcy Thompson's (1917, pp. 385–386) proposed insect wing study across four pre-registered clades spanning Diptera, Lepidoptera, and Hymenoptera (`drosophilidae_wings.py`, `morpho_butterflies.py`, `vespinae_wings.py`, `bee_wings.py`, `why_wings.py`; **Figure 3**):

- **Diptera — *Drosophilidae* Wings (**21,138 **Species Wings Across 115 Species +** 17,323 **$\mathbf{G}_w$ Wings +** 12,075 **M Wings, 12 Landmarks; Houle et al. 2017 *Nature*):** Across 115 drosophilid species, the 7 a-priori Thompson modes capture **71.3%** of between-species variance (96.5% in 7-D PCA, Hypothesis DW1 ✓), interior crossveins have significantly higher unexplained residual than outer margin landmarks (50.7% vs. 35.9%, Hypothesis DW2 ✓), phylogenetic signal is strong ($\hat{\lambda}_{\text{full}} = 0.806, \hat{\lambda}_{\text{T7}} = 0.602$, Hypothesis DW3 ✓), leave-one-species-out prediction across 83 tree species cuts error to **0.605**× consensus ($p = 2.8\times10^{-12}$, Hypothesis DW4 ✓), and standing additive genetic variance $\mathbf{G}_w$ aligns strongly with macroevolutionary divergence $\mathbf{D}$ ($|\cos\theta(\mathbf{G}_w, \mathbf{D})| = \mathbf{0.850}$).

- **Lepidoptera — *Morpho* Butterfly Forewings & Hindwings (**911 **Specimens Across 30 Species on Calibrated MCC Tree, 44 FW / 29 HW Landmarks; Chazot et al. 2021):** All 4/4 pre-registered hypotheses passed (Brier score **0.037**): Thompson-7 captures **49.0%** of forewing and **57.0%** of hindwing variance (Hypothesis MB1 ✓), Brownian Motion wins with $\hat{\lambda} = 0.877$ and 0.926 (Hypothesis MB2 ✓), LOO prediction cuts forewing error to **0.585**× consensus ($p = 4.7 \times 10^{-8}$, Hypothesis MB3 ✓), and within-species male→female sexual dimorphism is significantly more Thompsonian (**57.6%**) than between-species divergence (**49.0%**, Hypothesis MB4 ✓).

- **Shallow Divergence in Hornets (30 Myr,** 1,027 **Wings, 30 Species, 19 Landmarks; Perrard et al. 2014):** Within a hornet species, individual wing variation (**P**) is itself **54.0% Thompsonian**, proving that developmental growth variation follows smooth global modes. Yet between close hornet species (30 Myr), divergence drops to **38.9% Thompsonian** (below the 2,000-run **P**-drift simulation interval [44.8%, 62.2%]), sits 80.1° orthogonal to $\mathbf{p}_{\max}$, exhibits White Noise phylogenetic signal, and concentrates its residual in the central wing cells. Allometry explains 14.6% by ordinary regression ($p = 0.010$) but only 4.0% ($p = 0.719$) under phylogenetic GLS (reflecting the single *Vespa* vs. *Vespula* clade size contrast).

- **Deep Divergence in Bees (110 Myr,** 615 **Wings, 42 Tribes Across 10 Families + 2 Eocene Fossils, 18 Landmarks; De Meulemeester et al. 2014):** Over 110 Myr across bee tribes, the Thompson-7 share rises to **58.0%** (above the **P**-drift median of 51.5%), aligns with $\mathbf{p}_{\max}$ (38.4°, with 49% of among-tribe variance on the first 3 within-unit PCs vs. 9.4% random expectation), and enables phylogenetic prediction to beat both the consensus (0.85×) and copying the nearest relative (0.97×, $p = 0.019$).

- **Direct Confirmation of Hansen & Houle (2008) Per-Landmark Evolvability:** Across both hornets and bees, per-landmark digitizing noise does not predict where Thompson's global grid fails ($\rho = -0.22$ and -0.39, Hypothesis C1 ✓), whereas per-landmark among-taxon divergence strongly tracks per-landmark within-taxon variation ($\rho = \mathbf{0.62}, p = 0.005$ in hornets; $\rho = \mathbf{0.62}, p = 0.006$ in bees). Thus, local vein-cell differences between close species do not compound over macroevolutionary time, while global proportion shifts accumulate along the developmental channel over 100 Myr.

Data and Code Availability

All scripts, pre-registered hypothesis JSON files, and datasets are openly available in the *Infinite Morphospace* repository (https://github.com/fractastical/morpholib): `thompson/run_crab_carapace_paradox_audit.py`, `thompson/run_antscan_caste_allometry_tests.py`, `thompson/portfolio/groups/camponotus_ants.py`, `thompson/portfolio/groups/vespinae_wings.py`, `thompson/portfolio/groups/bee_wings.py`, and `thompson/portfolio/groups/why_wings.py`.

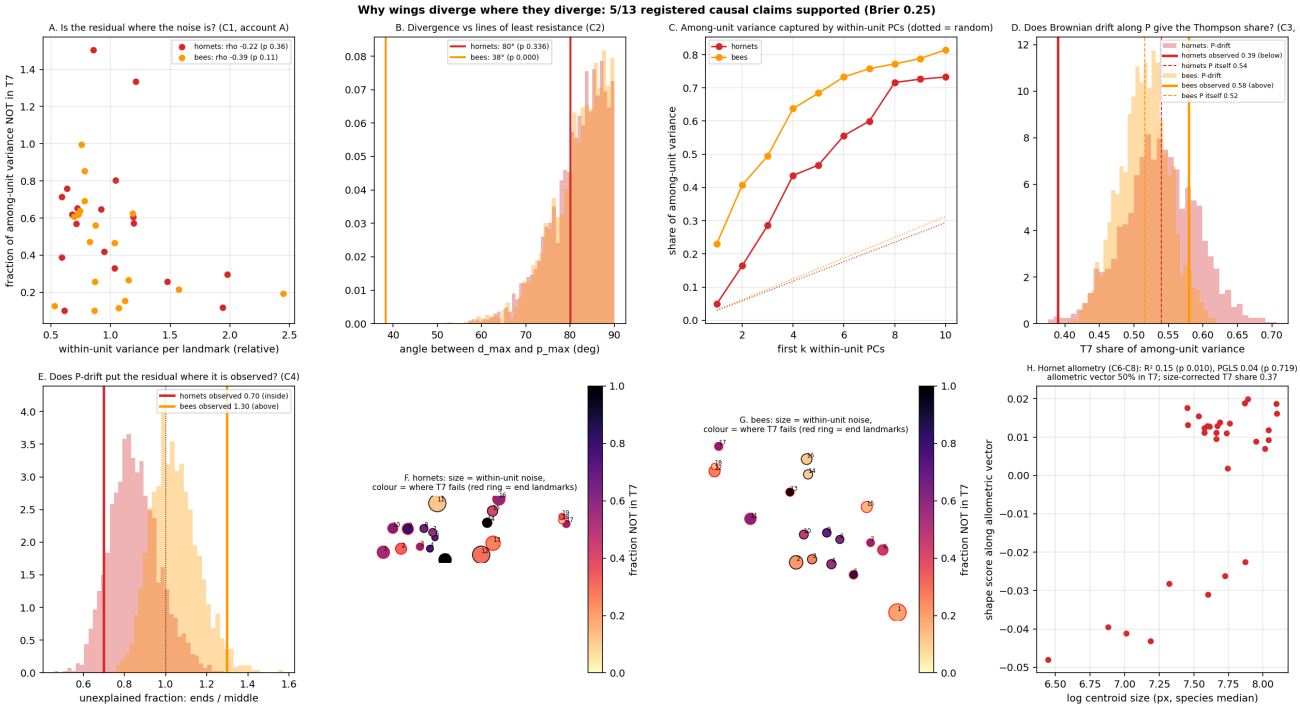


Figure 3: Causal Decomposition of Hymenopteran Wing Divergence Across Shallow (*Vespinae*, 30 Myr) and Deep (*Anthophila*, 110 Myr) Timescales. Within-unit variation **P** is 54% Thompsonian in hornets and 52% in bees, but shallow interspecific hornet divergence (38.9%) sits below the **P**-drift band and 80.1° orthogonal to $\mathbf{p}_{\max}$, whereas deep inter-tribal bee divergence (58.0%) aligns with $\mathbf{p}_{\max}$ (38.4°) at the upper edge of the drift band.

References

[1] Thompson, D. W. (1917). *On Growth and Form*. Cambridge University Press, Ch. XVII, pp. 385–386, 732–751.

[2] Scholtz, G., Knötel, D., & Baum, D. (2020). D'Arcy Thompson's 'on growth and form' and the computer-assisted-transformation of crab carapaces. *Zoomorphology*, 139, 299–314.

[3] Katzke, J., Economo, E. P., van de Kamp, T., et al. (2026). High-throughput 3D phenomics of ants (`ANTSCAN`). *Nature Methods*.

[4] Perrard, A., Baylac, M., Carpenter, J. M., & Villemant, C. (2014). Evolution of wing venation in hornets: why is the wing venation of Vespinae so variable? *PLoS ONE*, 9(4), e94198.

[5] De Meulemeester, T., Michez, D., Aytekin, A. M., & Danforth, B. N. (2014). Taxonomic affinity of bee forewings using geometric morphometrics. *PLoS ONE*, 9(5), e97154.

[6] Hansen, T. F., & Houle, D. (2008). Measuring and comparing evolvability and constraint in multivariate characters. *Journal of Evolutionary Biology*, 21(5), 1201–1219.